The $\mathrm{PSL}_3(4)$ incidence catalogue: classification, constructions and structural corrections

L100 source recovery

Abstract

We assemble a complete finite proof of the classification in the fixed 22-page Halle preprint of Gottschalk and Leemans. All subgroup anchors, primitive residue extensions and incidence isomorphism classes are covered. The resulting catalogue has 7, 7, 11 and 4 geometries in ranks 2, 3, 4 and 5, and none of higher rank. Literal projective constructions, doubling and truncation maps, residue data, stabilizer structures and complete automorphism and correlation groups are joined to the 29 printed rows. Several ancillary numerical, incidence and naming assertions require the explicit corrections below. The main classification is unchanged. Previously established classification inputs, algorithms and historical construction names are stated separately from the finite certificates.

1 Statement, conventions and fixed source

Let $G = \mathrm{PSL}_3(4)$ in its retained faithful 21-point realization, of order 20160. We consider finite coset geometries of rank at least two on which G acts flag-transitively. A geometry is *firm* if every panel has at least two objects, *residually connected* if every residue of rank at least two is connected, and *RWPRI* if the stabilizer of every flag acts primitively on at least one type of its positive-rank residue. The condition $(\mathrm{IP})_2$ says that every rank-two residue is a partial linear space or a generalized digon. No local two-transitivity condition is imposed in this classification.

We use right cosets with incidence given by nonempty intersection. Inversion converts the source's left-coset convention to this one. The standard firmness convention concerns panels, or equivalently nonmaximal flags in a geometry; the literal phrase "any flag" in the source definition must not be applied to a chamber itself. An isomorphism may permute entire types. The group $\mathrm{Aut}(\Gamma)$ fixes every type, whereas $\mathrm{Cor}(\Gamma)$ may permute them. An arbitrary automorphism of the uncolored incidence graph is not silently called a correlation.

Theorem 1. *Relative to the explicitly retained complete subgroup and exact computational inputs in Section 7, the firm, residually connected, G-flag-transitive RWPRI $(\mathrm{IP})_2$ geometries are precisely the 29 literal models in Table 1, up to incidence isomorphism with permutation of types. Their rank distribution is*

rank	2	3	4	5	≥ 6
number	7	7	11	4	0.

Every diagram, fusion count, structural group and construction clause has an explicit disposition below.

The source is the Halle report 99–30, *Geometries for the group* $\mathrm{PSL}(3, 4)$, by Harald Gottschalk and Dimitri Leemans [1]. Its exact 22-page file, page transcriptions and native diagrams are bound to the certificate. The 25-page journal edition has not been collated: a correction here is a statement about this fixed preprint. The original 55-clause reading register is preserved without alteration. The new register assigns every clause its proof, correction or retained antecedent; it does not overwrite the earlier reading-only record.

2 Complete primitive residue extension

The broad panel theorem [2] gives a finite reduction using primitivity alone. For a chamber stabilizer B and actual panel stabilizers $P_i > B$, put

$$K_\varnothing = B, \qquad K_S = \langle B, P_i : i \in S \rangle.$$

Recognition checks all intersections $K_S \cap K_T = K_{S \cap T}$, a maximal deletion $K_{S \setminus \{i\}} < K_S$ for each nonempty S, typed coset Helly for each subfamily of rank at least three, and the exact rank-two partial-linearity-or-digon test. The intersection and Helly conditions identify the residues with the corresponding coset geometries inside K_S. Maximality gives primitivity by the subgroup–block correspondence, even when the local action has a kernel. Connectedness gives generation of a residue stabilizer by its panel stabilizers. These observations establish both necessity and sufficiency, as proved in [2].

For completeness, take a target family of rank r. RWPRI supplies a maximal deletion H in its join K. By induction its rank-$(r-1)$ residue family has already occurred. The missing panel lies in K but outside H; adjoining it therefore generates K. All maximal overgroups and all eligible panels are tried. Stabilizer-orbit pruning uses actual automorphisms of the old tuple, and the final successive transporter/type-permutation test removes only equivalent families. No numerical signature is used as a sufficient isomorphism test.

The rank-two test retains every double coset. If $B = P \cap Q$ and $K = \langle P, Q \rangle$, the graph is a digon exactly when $|P||Q| = |K||B|$. Otherwise for each nontrivial BxB in $B \backslash Q / B$ the two neighbourhood sets

$$\{Qh : h \in P\}, \qquad \{Qhx : h \in P\}$$

must intersect in exactly one object. They already share Q; a second common object is precisely a quadrangle. This is the general test, not a shortcut valid only for two-transitive local actions.

The retained subgroup list has 95 G-conjugacy classes. Its permutation model is transported to the fixed realization by a checked bijective homomorphism with literal generator images. Complete local maximal inclusion data are transported by the same map. Every one of the 95 Borel anchors has now returned its final empty extension level. The final union has 145 full-group families up to inner conjugacy and type permutation: 21, 25, 63 and 36 in ranks 2–5. Earlier deadlines remain failures, not partial exhaustion certificates. The last anchor reused its completely saved levels and recomputed its unsaved next level from the beginning; its lost progress count was never treated as saved family data.

No higher family is omitted: every proper panel addition enlarges the join, so rank is bounded by the number of prime factors of $[G : B]$, with multiplicity. The acquired terminal empty-level checks, rather than agreement with the printed rank counts, establish exhaustion.

3 From specified actions to incidence classes

Inner equivalence does not identify all geometries. The complete automorphism group of G, of order 241920, is acquired with its actual generator-image maps. Its action and type permutations fuse the 145 families. The last 42 families were compared incrementally with the previous 103; previously completed comparisons were retained. The saved exact generator maps justify reuse of the already computed automorphism group and its cardinality. Computational faithfulness of the acquired nice representation is a stated input, not inferred from a cached size comparison.

Even full $\mathrm{Aut}(G)$ fusion alone would not classify bare incidence geometries: an incidence isomorphism need not normalize the chosen copy of G. The final comparison therefore uses the acquired palette construction [3]. Add one marker for every type, join it to exactly that type's objects, and distinguish object vertices from marker vertices by two colors. A color-preserving graph isomorphism restricts to precisely a type-permuting incidence isomorphism. Conversely

any such incidence isomorphism extends uniquely to the markers. Fixing the markers individually gives the type-preserving automorphism group. Thus the exact colored-graph computation supplies both bare equivalence and the full groups, without an uncolored shortcut.

All 29 resulting representatives have literal coset memberships, incidence matrices and generator actions. Each of the 145 families is joined to its representative; complete colored-graph comparisons separate the representatives. The rank distribution in Theorem 1 follows. The saved raw GAP output required a documented inverse of its line-continuation serialization; the derived files are bound to both the originals and 61 native round trips. A clean raw return alone is not used as evidence that malformed JSON was a scientific artifact.

4 The corrected catalogue

Table 1 uses the printed source-node order. Type sizes and panel orders therefore have an explicit node-by-node transport, including equal-sized types. The integer c is the number of inner conjugacy/type classes fused into the indicated incidence class; their sum is 145. The column $2T$ means that every panel action is two-transitive. The complete per-node flags are retained, including non-two-transitive primitive examples. Every bond, absent bond and printed ordered eccentricity/gonality triple is compared to the actual base residue, not merely to a diagram signature.

For the group columns put $A = \mathrm{Aut}(G)$ and let δ, σ, τ be diagonal, field and graph outer automorphisms. Their explicitly realized outer group is $S_3 \times C_2$, of order 12. Set

$$\Sigma = G \rtimes \langle \sigma \rangle, \quad T = G \rtimes \langle \tau \rangle, \quad U = G \rtimes \langle \sigma\tau \rangle,$$

$$\Gamma = G \rtimes \langle \delta, \sigma \rangle, \qquad V = G \rtimes \langle \sigma, \tau \rangle.$$

Here the notation denotes the actual extensions identified by the certificates, with the indicated outer images. Products $2 \times H$ in the table have a proved central complement, not just twice the order of H. The convention $\Sigma = P\Sigma L_3(4)$ is the extension by field automorphisms, as in the official Magma definition [11].

Table 1: All 29 corrected source rows.

Label	Type sizes	Panel orders	c	$2T$	Aut	Cor
2.1	$(21, 21)$	$(4, 4)$	1	yes	Γ	A
2.2	$(210, 56)$	$(14, 3)$	6	no	Σ	Σ
2.3	$(56, 280)$	$(1, 9)$	3	yes	V	V
2.4	$(120, 120)$	$(7, 7)$	3	yes	U	V
2.5	$(210, 120)$	$(6, 3)$	6	yes	Σ	Σ
2.6	$(280, 280)$	$(8, 8)$	1	yes	A	$2 \times A$
2.7	$(280, 1260)$	$(1, 8)$	1	yes	A	A
3.1	$(21, 210, 56)$	$(1, 4, 3)$	6	yes	Σ	Σ
3.2	$(56, 1260, 56)$	$(1, 8, 1)$	3	no	V	$2 \times V$
3.3	$(120, 210, 210)$	$(1, 2, 2)$	3	yes	Σ	V
3.4	$(56, 210, 210)$	$(1, 2, 2)$	3	yes	Σ	V
3.5	$(1120, 120, 120)$	$(6, 1, 1)$	3	no	U	V
3.6	$(56, 210, 56)$	$(3, 2, 3)$	6	yes	G	Σ
3.7	$(21, 105, 21)$	$(3, 4, 3)$	1	yes	Γ	A
4.1	$(21, 210, 56, 56)$	$(1, 1, 3, 3)$	6	yes	G	Σ
4.2	$(56, 1260, 21, 21)$	$(1, 4, 1, 1)$	3	no	Σ	V
4.3	$(120, 1260, 56, 56)$	$(1, 2, 1, 1)$	3	yes	Σ	V
4.4	$(21, 1260, 56, 56)$	$(1, 2, 1, 1)$	6	yes	Σ	Σ
4.5	$(56, 1260, 21, 21)$	$(1, 2, 1, 1)$	6	yes	Σ	Σ
4.6	$(120, 1260, 21, 56)$	$(1, 2, 1, 1)$	6	yes	Σ	Σ
4.7	$(120, 1260, 21, 21)$	$(1, 2, 1, 1)$	3	yes	Σ	V

Label	Type sizes	Panel orders	c	$2T$	Aut	Cor
4.8	$(56, 21, 21, 56)$	$(3, 1, 1, 3)$	6	yes	G	$2 \times \Sigma$
4.9	$(56, 21, 210, 56)$	$(1, 2, 2, 1)$	6	yes	G	Σ
4.10	$(21, 56, 56, 210)$	$(2, 1, 1, 2)$	12	yes	G	G
4.11	$(120, 120, 336, 336)$	$(1, 1, 2, 2)$	6	no	G	T
5.1	$(21, 56, 56, 210, 21)$	$(2, 1, 1, 1, 1)$	12	no	G	G
5.2	$(21, 56, 56, 210, 21)$	$(2, 1, 1, 1, 1)$	6	no	G	Σ
5.3	$(21, 56, 21, 56, 21)$	$(1, 1, 1, 1, 2)$	6	no	G	$2 \times \Sigma$
5.4	$(21, 56, 21, 56, 21)$	$(1, 1, 1, 1, 2)$	12	no	G	$2 \times G$

The exact structural table also records all 99 maximal parabolics and their 99 residue kernels, together with 29 Borels. Identical literal subsets reduce these positions to 77 atoms; 61 distinct actual parabolic/kernel pairs pay all quotient structures. The proof [9] uses faithful geometric actions, actual normal subgroups and complements, and complete normal forms. It never identifies a group solely from its cardinality. For example a hyperoval stabilizer has the entire alternating action on six points; a Baer stabilizer acts as the full seven-point Fano automorphism group; point and line stabilizers have actual translation radical 2^4 and Levi complement A_5. Small atoms are paid by cyclic, dihedral, quaternion and elementary-abelian semidirect normal forms.

Every full automorphism/correlation group is likewise identified inside the acquired complete palette group. Coordinate and duality actions are verified on every object and incidence, their actual outer images are recorded, and each extra central factor is split by an explicit binary adjustment of lifts [8]. Equality with the complete palette group uses exact orders after proving inclusion and the action map; equality of orders alone does not define the groups.

5 Constructions and their exact source joins

The two rank-five projective constructions use lines, points, two distinct hyperoval orbits and point pairs in $\mathrm{PG}(2, 4)$. The accepted coordinate-to-permutation map is checked on the actual generators. All ten pairwise incidence rules are tested, every partial flag extends to a chamber, and the chamber orbit has length 20160. Hence the action on chambers is regular, in particular kernel-free. Base-residue connectivity, primitivity and $(\mathrm{IP})_2$ checks transport to all residues by flag transitivity. All six ordered choices of distinct hyperoval classes have explicit coordinate/duality maps, checked on every relevant object type and incidence [4]. The single switched line–hyperoval rule gives Geometry 5.2.

For Geometry 5.1 the 20 extension numbers of Table 1 and all flag orders agree with the source. Lemmas 4.1–4.3 are paid by the complete flag, residue and action arguments. The supplement [10] identifies all 31 nonempty flag stabilizers, rather than inferring the intermediate names from those orders. In particular it proves $H_{123} \cong D_{10}$ and the actual fixed-point-free split $H_{25} \cong 2^4 : 3$. The source type order is fixed before these intersections are formed.

The Baer construction of 3.3 has 120 subplanes in a selected orbit, 210 point pairs and 210 line pairs. Its literal printed relation between the latter types needs the correction in Section 6. The corrected three-type incidence structure has explicit bijections to its classified coset representative; all incidences and actions are checked [5].

The six butterfly recipes 4.2–4.7 are supplied on all three Baer orbits, with 1260 ordered butterflies per orbit. Under the explicit all-four-cross-intersections convention, each positive pole has exactly one opposite pole. The stabilizer is proved to be $C_2 \times D_8$ by actual sixteen-word normal forms. The two hyperovals at each pole, their signs and central-line data are retained literally. Eighteen full object/incidence maps join these recipes to their six catalogue rows [7].

For 4.11 the two Baer classes are distinct. The relative oval and dual-oval orbit choice is explicitly supplied; it is not inferred from the printed clauses, which do not specify it. Six actual choices have complete four-type maps. Across these constructions the certificates check

all 1,935,360 coset members, 10,679,394 cross-type incidence pairs and 61,824 generator actions. These finite verification totals describe the supplied maps, not new classification counts.

Four doubles have complete literal isomorphisms:

$$2.7 \longmapsto 2.6, \qquad 4.1 \longmapsto 4.8, \qquad 5.2 \longmapsto 5.3, \qquad 5.1 \longmapsto 5.4.$$

The rank-two case has its own elementary incidence proof. The quoted general doubling antecedent requires rank at least three and its weaker $I_{1,2}$ shadow condition; neither hypothesis is silently changed. Sixteen new truncation maps and eight acquired maps pay the stated truncation links after the correction below. The literal four doubles, maps and fusion counts are in [6]. Earlier named constructions cited only by reference in the source remain attributed antecedents, as specified in Section 7.

6 Ancillary corrections and qualifications

These corrections leave the principal list of incidence geometries unchanged. Each is attached to a fixed source clause and exact literal countercheck, not to an uncollated journal edition.

1. In the diagrams 5.1 and 5.2 the 210-object node has panel order 1, not the printed 2. The complete construction flag counts and source-node transports agree on this correction.

2. The printed 3.6 truncation link from 5.3 is impossible: 5.3 has no type of 210 objects. The corrected 5.2 link has an explicit incidence isomorphism.

3. In the printed recipe for 3.3, the point-pair/line-pair relation has degree 18 and admits a flag with no extension to the Baer type. Requiring a common Baer subplane in the selected orbit gives degree 6 and the certified construction. This concerns the ancillary recipe, not nonexistence of Geometry 3.3.

4. Under the stated field convention for $P\Sigma L$, the typed group of 3.5 is U, and the correlation group of 4.11 is T, rather than the printed Σ. These are not relabelings based on orders. The three outer involution cosets contain 120, 280 and 336 involutions, respectively; adding the 315 inner involutions separates the three abstract extensions. Literal outer images and full group equalities are also certified.

5. In diagram 2.3 the actual kernel $3^2 : C_4$ in $3^2 : Q_8$ has index two and no involution outside it. Thus it has no complement. Under the usual split-colon convention the outer colon in $(3^2 : 4) : 2$ must be replaced by an extension dot.

Two further qualifications concern how a recipe is specified. On p. 4 the two displayed corresponding cross-intersections for butterflies alone do not force the advertised unique completion; explicit alternative completions are retained. The certified recipe imposes all four cross-intersections. This is a statement about the displayed conditions, not an adjudication of every external butterfly convention. On pp. 18–19 the oval/dual-oval relative orbit choice is supplied explicitly, as noted above.

The generic residue-kernel dot defined on p. 3 does not assert nonsplitting. Accordingly central 2.2^3 extensions are proved nonsplit separately, and unparenthesized $2^2 \times 2^2 : L$ displays are interpreted as the actual action on the entire radical; they do not imply an unproved central factor or invariant complement. The earlier working allegation of a missing horizontal bond in 3.5 was our own transcription error and is withdrawn. The native source has that bond; it is not among the source corrections.

7 Retained inputs, attribution and verification

The proof has the following explicit antecedent boundary.

- The complete TomLib subgroup classification for $L_3(4)$, its literal transported realization, and complete local maximal inclusion data are retained. GAP's exact subgroup, homomorphism, transporter and automorphism algorithms, and GRAPE/nauty's complete colored-graph computations are computational premises. Independent small-model checks and positive witness maps do not reprove these classification tables or software algorithms.

- The source's quoted Tits connectivity criterion and Buekenhout–Hermand coset-Helly criterion remain attributed with their hypotheses. The actual general recognition and augmentation arguments used here are the complete acquired proofs in [2]. We do not turn the source's short strongly-Boolean sentence into a stronger theorem lacking its setting.

- Classical terminology and uniqueness of $\mathrm{PG}(2,4)$ are retained. Every concrete point, line, hyperoval, Baer set and butterfly used by the current constructions is separately bound to its literal finite certificate. The general rank-at-least-three doubling theorem is retained exactly as quoted, with its $I_{1,2}$ condition; all four finite applications have direct maps.

- The source's retrospective thesis reduction and earlier Cayley/Magma computations are historical attribution. The present proof uses the complete acquired panel computation instead of claiming to rerun those programs. Earlier construction references, including the Gewirtz and unital graph names attached to 2.3 and 2.7, are retained as attributed antecedents. All 29 current coset models and all newly prescribed source recipes are supplied; we do not claim newly to inspect every cited primary article or recertify each external naming identification.

The machine-readable register preserves all 55 original clauses, 47 bibliographic attributions and the exact fixed-edition scope. All 29 diagram clauses acquire numerical, structural and construction routes, with the six correction records above (the two group-name sites are separate records). The source contains no named unresolved research question; omitted details and classification obligations are not relabeled as new questions. No present-day conjecture-status claim, Lean formalization, measurement or novelty claim is made.

The principal checker verifies every acquired input hash, the exact 95-anchor/145-family/29-model joins, all 55 source dispositions, the complete native manuscript inputs and damaged-interface controls. It does not repeat the acquired group or graph censuses. The complete proofs named below and their separate review records are included in the acquisition closure. Authorship of this synthesis and the component-specific review roles are recorded explicitly; root's previews are not substituted for final distinct acceptance.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* $\mathrm{PSL}(3,4)$, Halle Reports on Algebra and Geometry 99–30, fixed 22-page preprint.

[2] *Primitive residue extension for a fixed finite acting group*, acquired broad preparation v2; complete target, continuation and fusion certificates, L100 finite archive.

[3] *Literal palette isomorphisms and exact serialization repair*, acquired L100 incidence archive, v1.

[4] *The projective constructions for the* $\mathrm{PSL}_3(4)$ *catalogue*, acquired construction component v1.

[5] *The Baer-subplane construction and its printed-rule correction*, acquired L100 component v1.

[6] *Doubling, truncation and fusion joins*, acquired L100 component v1.

[7] *Ordered butterflies, ovals and their incidence constructions*, acquired L100 component v2.

[8] *Automorphism and correlation group structures in the* $\mathrm{PSL}_3(4)$ *catalogue*, acquired L100 component v1.

[9] *Parabolics, Borels and residue kernels in the* $\mathrm{PSL}_3(4)$ *catalogue*, acquired L100 component v2.

[10] *Intermediate flag stabilizers in Geometry 5.1*, acquired L100 supplement v1.

[11] Magma Handbook, projective classical groups, definition of $P\Sigma L$: `https://magma.maths.usyd.edu.au/magma/handbook/text/713`. Exact inspected notation record retained in [8].

Primitive residues in rank two

Human Compression project

Working proof, 14 October 2026

Let G be finite. We use right cosets, with incidence given by nonempty intersection. Firmness means that every panel lies in at least two chambers. For a rank-two geometry, residual weak primitivity requires a primitive G-action on at least one type and primitive actions of both vertex stabilizers on their respective neighbours. Thus it includes conditions on the rank-one residues. No double transitivity is assumed.

Theorem 1. *All connected, firm, flag-transitive, residually weakly primitive rank-two geometries for G satisfying the intersection property are obtained by the following construction.*

1. *Choose one representative A of each conjugacy class of maximal proper subgroups of G.*
2. *Choose one representative H of each A-conjugacy class of maximal proper subgroups of A.*
3. *Take every actual subgroup B with $H < B < G$, retaining it precisely when $A \cap B = H$ and H is maximal in B.*
4. *Retain the pairs passing Lemma 2. Identify pairs under automorphisms of G and interchange of their entries.*

The final identification classifies the specified G-actions. Distinct resulting classes may require a further comparison if one wishes to forget the specified action and classify abstract incidence geometries.

Proof. Choose an incident pair in such a geometry. Its stabilizers A, B identify the geometry with the incidence structure of their cosets, by flag transitivity. Interchange the types so that A is maximal in G, and conjugate to the chosen representative. The common stabilizer $H = A \cap B$ is maximal in both A and B: in a transitive permutation action, primitivity is equivalent to maximality of a point stabilizer. Conjugating within A now puts H in the second list. The resulting B is one of the actual overgroups required in the third step. This proves completeness before the incidence test and equivalence comparison.

Conversely, the two proper containments $H < A, B$ give firmness. Since $H < B$ and $A \cap B = H$, the subgroup B is not contained in A. Maximality of A therefore gives $\langle A, B \rangle = G$, which proves connectivity. Every incident pair of cosets contains a common element, so it is a translate of (A, B); hence the action is flag-transitive. The three maximality conditions give precisely the primitive actions required above. These are all the nonempty residues that need checking in rank two. Lemma 2 supplies the intersection property. An automorphism carrying one pair to another, possibly interchanging its entries, induces the corresponding incidence isomorphism. □

Lemma 2. *For a pair obtained in the third step, let T_A, T_B be right transversals of H in A, B. For each $a \in T_A \setminus H$, put*

$$c(a) = \big|\{b \in T_B : \text{ some } d \in T_B \text{ has } b(da)^{-1} \in A\}\big|.$$

The intersection property holds exactly when all values $c(a)$ equal 1, or when $[B : H] = [G : A]$.

Proof. The neighbours of B are Ab for $b \in T_B$, while those of Ba are Ada for $d \in T_B$. Both lists have distinct entries because $A \cap B = H$. Equality of these two right cosets is exactly the displayed membership test, so $c(a)$ counts their common neighbours. The cosets Ba with $a \in T_A$ exhaust the neighbours of A. The group A is transitive on these neighbours; consequently every pair of distinct neighbours of A can be moved to (B, Ba) for one of the tested cosets. Transitivity on vertices of the first type now shows that the condition $c(a) = 1$ for all a is equivalent to having no quadrangle anywhere in the incidence graph. This is the partial linear space condition.

The other allowed possibility is a generalized digon, namely a complete bipartite geometry. A vertex of the second type has $[B : H]$ neighbours among $[G : A]$ vertices of the first type. By transitivity, completeness is therefore equivalent to equality of these numbers. The two alternatives are exactly the required intersection property. □

Application to $\mathrm{PSL}(3, 4)$

The computation uses the degree-21 permutation representation supplied by GAP. It acquires the complete maximal-subgroup algorithms and data for G, its maximal subgroups, and the intermediate groups encountered during the overgroup search. These subgroup classifications are retained inputs; the argument above does not reprove them. In particular, the routine for intermediate subgroups must return actual subgroups, rather than only conjugacy representatives.

There are nine conjugacy classes of maximal subgroups. Their local lists give 42 pairs (A, H). The third step yields 60 pairs (A, B), of which 31 pass the incidence test. They give the following seven classes. The source labels refer to the complete Halle preprint of Gottschalk and Leemans, *Geometries for the group PSL(3,4)*, Report 99–30 (1999), pp. 7–8; its later journal edition has not been collated here.

Source	$\lvert A\rvert$	$\lvert B\rvert$	$\lvert H\rvert$	$[A : H]$	$[B : H]$	$(2T)_1$
2.1	960	960	192	5	5	yes
2.2	360	96	24	15	4	no
2.3	360	72	36	10	2	yes
2.4	168	168	21	8	8	yes
2.5	168	96	24	7	4	yes
2.6	72	72	8	9	9	yes
2.7	72	16	8	9	2	yes

Both local actions are doubly transitive in six of the seven classes. Class 2.2 has a primitive action of degree 15 whose point stabilizer has order 24; it cannot be doubly transitive, since 14 does not divide 24. Its other local action, of degree 4, is doubly transitive. These are outputs of the computation: the search includes every primitive local action and rejects pairs only by the stated tests. The seven unordered pairs of parabolic orders are distinct. Since the two type sizes are $[G : A]$ and $[G : B]$, the seven representatives are also inequivalent as abstract incidence geometries, even allowing interchange of types. Explicit automorphisms identifying the other surviving pairs are supplied separately as images of generators.

The coset correspondence and the stabilizer criterion for primitivity are classical. The present argument organizes these inputs into the same finite construction used for doubly transitive residues, with the incidence test extended to all local pairs. No claim of historical novelty for this enumeration principle is made.

Primitive residue extension for a fixed finite acting group

Private L100 applicability component

This note specializes the previously accepted Topic D arguments: anchored Helly, panel recognition, subgroup transport and maximal augmentation. The printed augmentation theorem there imposes local two-transitivity. Here that extra restriction is removed explicitly; primitivity alone is used. The exact general double-coset test already present in the phase-six and phase-seven programs is retained. No new enumeration of the $\mathrm{PSL}(3,4)$ geometries is asserted in this note.

All cosets below are right cosets, with incidence defined by nonempty intersection. Inverting every group element converts the source's left-coset convention to this convention and preserves incidence. Geometries are firm when each panel has at least two objects, residually connected, and flag-transitive under the specified acting group. RWPRI means that in each positive-rank residue the specified residue stabilizer acts primitively on at least one type. The rank-two axiom $(\mathrm{IP})_2$ means that each rank-two residue is a partial linear space or a generalized digon. A local action may have a kernel; subgroup maximality still characterizes its primitivity.

Theorem 1 (Broad primitive panel recognition)**.** *Fix a finite group G, $B \le G$ and distinct subgroups $P_i > B$ indexed by a finite set I. Put $K_\varnothing = B$ and $K_S = \langle B, P_i : i \in S\rangle$. The family defines a firm, flag-transitive, residually connected RWPRI $(\mathrm{IP})_2$ geometry for its acting group K_I, with chamber stabilizer B, precisely when:*

1. *$K_S \cap K_T = K_{S\cap T}$ for all $S, T \subseteq I$;*
2. *for every nonempty S, some $K_{S\setminus\{i\}}$ is maximal proper in K_S;*
3. *for every $|S| \ge 3$, the deletion parabolics $(K_{S\setminus\{i\}})_{i\in S}$ have typed coset Helly;*
4. *for every two-element S, the corresponding coset graph is a partial linear space or a generalized digon.*

Conversely every geometry with these properties arises from its actual panel stabilizers in this way.

Proof. The first condition, together with proper panels and the top Helly condition, is the accepted panel recognition theorem. It identifies the stabilizer of the complementary base flag with K_S, and identifies each residue with the corresponding coset geometry inside K_S. The singleton cases of the second condition give proper maximal inclusions $B < P_i$ and hence firmness. All other cases of that condition give exactly RWPRI, by the block–subgroup correspondence for a transitive coset action; its kernel does not affect that correspondence. The fourth condition gives exactly $(\mathrm{IP})_2$ in the identified residues.

For necessity, use actual flag stabilizers. Their intersections give the first condition. The stabilizer of a residue is generated by its panel stabilizers: in rank two this is the coset-path connectedness argument; in higher rank, connectedness and induction first give generation by the maximal parabolics, then by their panel stabilizers. Equivalently one can lift an incidence path to adjacent chambers, using the already connected lower-rank residues. This retains the common chamber stabilizer B at every step. Flag transitivity gives typed Helly, and every residue inherits all four conditions. Thus no local two-transitivity premise is used. □

Lemma 1 (The general rank-two test)**.** *Let $B = P \cap Q$ and $K = \langle P, Q \rangle$. The coset graph is a digon if and only if $|P||Q| = |K||B|$. Otherwise it is a partial linear space if and only if, for every nontrivial double coset BxB of B in Q,*

$$|\{Qh : h \in P\} \cap \{Qhx : h \in P\}| = 1.$$

Proof. The neighbours of P are the Qh with $h \in P$; their number is $[P : B]$. This equals the entire number $[K : Q]$ of opposite objects precisely in the displayed digon case. Otherwise a quadrangle exists precisely when two distinct P-type objects have two common neighbours. Translate one common neighbour to Q, and then translate one of the two objects to P by an element of Q. The other is Px for $x \in Q \setminus B$. The remaining stabilizer of the base incident pair is B, and its orbits on these objects are the double cosets $B\backslash Q/B$. The two neighbour sets are the displayed sets. They already share Q, so the required bound of at most one is exactly equality with one. This treats every double coset, without a two-transitivity shortcut. □

Theorem 2 (Complete primitive augmentation)**.** *Assume representatives of every G-conjugacy class of subgroups are supplied, together with representatives of every K-conjugacy class of maximal subgroups for each representative K. For each Borel representative B, let $\mathcal{P}(B)$ contain* all *actual overgroups P in G with B maximal proper in P. Apply the phase-seven augmentation with `REQUIRE_2T=false`, its general rank-two test, and all four recognition conditions. If every Borel run reaches its certified empty next level, the outputs with join G are exactly the requested geometries, represented as subgroup families, modulo inner conjugacy and permutation of types.*

Proof. UD.Transport enumerates all maximal overgroups of any given subgroup from the retained inclusion data. There is no local-action filter in $\mathcal{P}(B)$. In a target of rank r, RWPRI at the top supplies a maximal deletion H in the target join K. Its rank-$(r-1)$ panel family is a residue satisfying the same conditions, so induction reaches it. Its missing panel P lies in K but not in H; since H is maximal, adjoining P has join exactly K. Every such maximal overgroup and every such panel is tried. The pointwise stabilizer of the old tuple in the image of $N_G(B)$ acts on these extensions; taking its orbits loses no family. Using this possibly smaller stabilizer instead of the setwise stabilizer only leaves duplicates. The program's final successive-transporter test and type permutations remove exactly those duplicates. Its order and pair-join signatures are necessary invariants, never sufficient equivalence tests. Conversely each accepted family passes the exact recognition conditions.

The intersection condition forces every new panel to enlarge the join: $P_i \leq K_S$ for $i \notin S$ would imply $P_i = P_i \cap K_S = B$. Consequently the rank is at most $\Omega([G : B])$, counting prime factors with multiplicity. The extra next level must therefore be empty. A timeout, missing Borel, unfinished anchor, or numeric cap before this terminal test is not an exhaustion certificate. Finally every Borel is conjugate to an input representative, and conjugating a family also conjugates its Borel. □

Exact implementation boundary. The retained engine is `phase7_sporadic_search.g`. Its seven top-level function definitions, from `IsMaxPair` through `RunBorel`, and its maximal-inclusion initializer are extracted byte-for-byte; the legacy file is unchanged. The only new interface supplies the complete explicit group and subgroup representatives and fixes the broad-mode flag. `FlagTransitive` uses the anchored Helly test: for each distinguished parabolic it intersects the sets of cosets meeting the other base parabolics and compares their cardinality with the contained set of cosets meeting their intersection. All subfamilies of size at least three are checked. `ValidPanels` checks every subgroup intersection and every maximal deletion. `RankTwoAdmissible` additionally checks the two-deletion RWPRI condition. `IsMaxPair` is called only for known inclusions. Prime index is an exact sufficient shortcut; all other branches use maximal-subgroup transport or the actual primitive coset action. The union of maximal subgroup orders is only a necessary pruning test.

L100 retained inputs and equivalence. The installed TomLib 1.2.11 table named `L3(4)` has 95 subgroup classes. Its underlying degree 21 permutation group has order 20160 but is not literally the retained named action. A separately checked homomorphism given by images of its two generators is bijective onto the retained group. Transporting the complete representative and inclusion lists by that map preserves all conditions. TomLib's complete subgroup classification and GAP's exact maximal-subgroup, transporter and group algorithms remain named retained premises, not conclusions of the small controls. No claim is based merely on equal orders or names.

The first census would produce inner-conjugacy/type classes. A later exact $\mathrm{Aut}(G)$ fusion would classify the specified-action objects up to automorphism of G and type permutation. This alone is not asserted to classify bare incidence geometries: an arbitrary incidence isomorphism need not normalize the specified subgroup G. An additional faithful incidence-isomorphism comparison or a proved uniqueness-of-G argument would be required for that assertion. Full automorphism and correlation groups, displayed kernel/group names, all remaining constructions, and the whole source contract are separate obligations.

Bounded controls and attribution. Literal multiplication-table controls independently enumerate all subgroups and all panel subsets in $S_3, C_2^2, D_8, A_4, C_3, C_3^2, C_2^3$. They compare every partial and full family, all Borel representatives, terminal empty levels, and normalizer/type orbits with the unchanged engine. They use direct coset incidence for Helly and $(\mathrm{IP})_2$, not the extension recurrence. The cyclic-three controls exhibit primitive non-two-transitive panels; the elementary-abelian order-eight control includes rank three. These controls are finite checks of the new interface, not a proof of TomLib completeness or a target census. The common recognition, transporter and extension arguments are attributed to the accepted Topic D work and its classical coset-action antecedents. The independent preliminary read of the broad specialization is source/interface scrutiny only. No novelty, measurement, formalization or source-disposition credit is claimed here.

Two literal projective-plane constructions for $\mathrm{PSL}_3(4)$

A bounded recovery of constructions 5.1 and 5.2 in Gottschalk–Leemans

The reference edition is the complete 22-page Halle preprint [1], not an uncollated later journal edition. We recover the two incidence constructions on its physical pages 14–16, Lemmas 4.1–4.3 and Table 1. The finite certificate also treats the specified truncations below. This is an existence and incidence result, not an exhaustion of higher-rank geometries. Automorphism and correlation groups, conjugacy-fusion counts, doubling and the other constructions are not asserted here.

The actual finite model. Put $F = \mathbb{F}_2[w]/(w^2 + w + 1)$ and label its elements $0, 1, 2 = w, 3 = w+1$. List the 21 projective points by lexicographically sorting the nonzero column vectors whose first nonzero coordinate is 1. Lines are the kernels of the 21 normalized nonzero linear forms. A hyperoval is a six-subset with no three collinear points. Let $G = \mathrm{PSL}_3(4)$ in its point action.

We use the two actual degree-21 generators retained in the independently reviewed rank-two component. No equality of independently ordered labels is presumed. A five-point orbit of its order-960 line stabilizer gives a line; the orbit of that set gives all 21 lines. The certificate verifies the coordinate-to-retained label map

$$(6, 2, 10, 14, 18, 1, 7, 8, 9, 3, 11, 16, 21, 4, 19, 12, 17, 5, 15, 20, 13).$$

It checks every line under this bijection. In these coordinates the actual two retained generators are represented, respectively, by

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & w & 0 \\ 0 & 0 & w+1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}. \tag{1}$$

Both determinants are 1; all 21 images are checked, not merely a projective frame. Closure of the two literal permutations has 20,160 elements. This is also $|\mathrm{SL}_3(4)|/3 = 4^3(4^3 - 1)(4^2 - 1)/3$, so the realization is the full named projective special linear group. The retained named-model identification and finite-field conventions are explicit inputs; the literal matrix and permutation checks additionally verify their use here.

Testing all $\binom{21}{6} = 54,264$ subsets gives 168 hyperovals, in three G-orbits of 56. Fix distinct orbits $\mathcal{H}_3, \mathcal{H}_4$. The five types are lines, points, $\mathcal{H}_3$, $\mathcal{H}_4$, and unordered point pairs. The type sizes are $(21, 21, 56, 56, 210)$. For $h \in \mathcal{H}_3$, $k \in \mathcal{H}_4$ and Q a point pair, define the ten relations by

	p	h	k	Q
ℓ	$p \notin \ell$	$\ell \cap h = \varnothing$	$\ell \cap k \neq \varnothing$	$\ell \cap Q = \varnothing$
p		$p \in h$	$p \in k$	$p \in Q$
h			$\lvert h \cap k \rvert = 3$	$Q \subset h$
k				$Q \subset k$.

(2)

This is $\Gamma_{5.1}$. For $\Gamma_{5.2}$ replace only $\ell \cap k \neq \varnothing$ by $\ell \cap k = \varnothing$. These are the literal rules in the source; in particular line–point incidence is nonincidence in the underlying projective plane.

Theorem 1. *Each of these two incidence systems is a firm rank-five geometry with 20,160 chambers. The actual G acts regularly on chambers and transitively on flags of every type. Each geometry is residually connected, residually weakly primitive, and satisfies* IP_2. *Its rank-one residue*

double-transitivity flags, in type order, are $(\text{false}, \text{true}, \text{true}, \text{true}, \text{true})$. *For* $\Gamma_{5.1}$ *all twenty one-type extension numbers in source Table 1 are those displayed in Table 1. The choice of the ordered pair of distinct hyperoval orbits does not change either geometry up to a type-preserving isomorphism normalizing the specified* G*-action.*

Exact finite verification and its completeness. The accompanying program uses only integer masks, permutations and finite field arithmetic after exporting the fixed input generators and line orbit. For a prescribed ordered type set I, it recursively chooses an object of each type, intersecting the remaining allowable sets with its adjacency row. Induction on $|I|$ proves that the returned tuples are precisely all flags of those types, without repetition. This procedure is applied to every subset of the five types, including the empty subset.

Let $\mathcal{C}$ be the resulting complete set of full flags and choose $C \in \mathcal{C}$. Applying every one of the 20,160 literal elements of G to C gives 20,160 distinct tuples, exactly $\mathcal{C}$. All generator actions preserve every object set and all ten incidence relations. Thus this is a genuine regular chamber action; its kernel is trivial, since an element in the kernel fixes C. Independently, for every type subset I the complete flag list is compared with the projection of $\mathcal{C}$ onto I. Equality proves that every partial flag extends to a chamber; chamber transitivity alone is not substituted for this check. It then also proves flag transitivity.

For each $F \subseteq C$, the residue objects of type i are computed as the intersection of the adjacency rows of the objects of F. Its action uses precisely those literal elements of G fixing every object of F. No permutation-group quotient or replacement action is used. The incidence graph of each residue of rank at least two is connected by direct traversal. Each panel has at least two extensions, proving firmness under the standard panel convention; a chamber itself is not required to have two extensions. Because every flag is conjugate to the corresponding subflag of C, these are checks of all residues, including those based at flags not chosen in the calculation.

Lemma 1. *Let* K *act transitively on a finite set* X*, with* $x \in X$*. For one representative* y *of each* K_x*-orbit on* $X \setminus \{x\}$*, form the undirected graph with edges* $\{x^g, y^g\}$ *for all* $g \in K$*. The action is primitive if and only if all these graphs are connected.*

Proof. Connected components of any such graph are a K-invariant equivalence relation, so a disconnected graph contradicts primitivity. Conversely, if a nontrivial block contains x and $y \neq x$, all edges in that pair orbit lie within blocks; its graph cannot be connected. Choosing another point in the same K_x-orbit gives the same graph. □

The certificate applies this lemma to the actual actions of each G_F on each remaining type. Every residue has at least one primitive type action, precisely the weak primitivity required by RWPRI. Double transitivity is recorded separately by the orbits of an actual point stabilizer; it is never a filter. For a rank-two residue, the program checks that the graph is complete bipartite or that each pair on one side has at most one common neighbour. The latter is symmetric (a pair of common neighbours would give a quadrangle) and proves the partial-linear condition IP_2. These checks use every pair of local objects.

The finite output retains all 32 flag counts, all actual stabilizer orders, all 31 positive-rank residue records, every local primitivity and double-transitivity flag, and every rank-two test for each model. In both models the panel orders in type order are $(2, 1, 1, 1, 1)$; equivalently the five panel sizes are $(3, 2, 2, 2, 2)$. Their chamber stabilizers are trivial. The first panel action is the regular cyclic group of order 3 and is not doubly transitive; the other four panel actions have degree 2. Table 1 is computed directly at the corresponding base triple flags; the already verified flag action transports it to every triple flag.

The six choices of hyperoval orbits. The coordinate maps $\mathrm{diag}(w, 1, 1)$ and $(x, y, z) \mapsto (x^2, y^2, z^2)$ preserve the plane and normalize the actual group: conjugates of both retained generators are checked in the complete literal G. Their permutations of the three hyperoval orbits

fixed types	added type	number	fixed types	added type	number
123	4	5	123	5	5
124	3	3	124	5	3
125	3	2	125	4	2
134	2	3	134	5	3
135	2	2	135	4	2
145	2	2	145	3	2
234	1	3	234	5	2
235	1	6	235	4	4
245	1	6	245	3	4
345	1	3	345	2	2

Table 1: Exact source Table 1 extension numbers for $\Gamma_{5.1}$.

are a 3-cycle and a transposition. They therefore act transitively on the six ordered pairs of distinct orbits. For each of these six pairs, for both models, and for each of the two maps, the certificate stores the resulting bijections on all five object types and checks every incidence and nonincidence. These 24 complete maps pay the choice-independence assertion. They are not a computation of the full automorphism group.

The specified truncations. The literal restrictions in Table 2 have the indicated source profiles. Here the type numbers always refer to (2), B is the chamber stabilizer in the specified G, and panel orders are listed in increasing retained-type order. For each row the program again enumerates every partial flag and chamber, proves flag transitivity, and verifies every base residue, firmness, RWPRI and IP_2. It checks that projection of all rank-five chambers is exactly the truncated chamber set.

source label	parent	retained types	panel orders	$\lvert B\rvert$
4.1	5.1, 5.2	2345	$(1,3,3,1)$	3
4.10	5.1	1345	$(2,1,1,2)$	2
3.1	5.1, 5.2	235	$(1,3,4)$	12
3.6	5.1	345	$(3,3,2)$	6
2.2	5.1, 5.2	35	$(3,14)$	24

Table 2: Eight checked literal truncation routes.

The object maps are the identity on each retained type; incidence is literally restricted, rather than inferred from the numerical profile. The two 4.1 restrictions and the two 3.1 restrictions are therefore identical incidence models: they omit the only relation changed between 5.1 and 5.2. In particular the former also restricts to the latter and to 3.6. The 4.10 restriction likewise restricts to 3.6. These maps establish the advertised construction routes in this fixed edition. They do not identify an arbitrary geometry from its diagram alone. No further independent realization from the cited earlier papers is being asserted isomorphic without a supplied map.

The rank-two 2.2 restriction has local degrees $(15,4)$ with parabolic orders $(360,96)$ and intersection order 24. Its actual local double-transitivity flags are $(\mathrm{false},\mathrm{true})$, consistent with the corrected accepted rank-two component: $14 \nmid 24$. Thus the local-2T distinction is preserved through the higher-rank construction.

Scope and attribution. The constructions and their intended geometric properties are due to Gottschalk and Leemans and to the predecessors explicitly cited in their source. The new material here is a transparent finite certificate of those specified constructions, together with the exact

action and source joins. The accepted rank-two model is acquired unchanged. Complete higher-rank classification, other incidence constructions, doubling, full group names and kernels beyond the orders stated above, full automorphism and correlation groups, and fusion multiplicities remain separate obligations. No novelty, whole-source, question, Lean or measurement credit is claimed.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL(3, 4), Halle Reports on Algebra and Geometry 99–30, fixed 22-page preprint; physical pp. 8–16 for the source routes used here.

[2] *Primitive residues for* $\mathrm{PSL}_3(4)$, accepted L100 rank-two component, edition 2, with distinct review and root adoption; exact immutable manifests are listed in the certificate dependencies.

Literal diagram and construction joins for the $\mathrm{PSL}_3(4)$ catalogue

Finite source-recovery component

The fixed source is Gottschalk and Leemans, *Geometries for the group* PSL(3, 4), Halle Reports on Algebra and Geometry 99–30, the complete 22-page institutional preprint. Its PostScript and derived PDF are retained exactly. The 25-page journal edition, European Journal of Combinatorics 24 (2003), 267–291, has not been collated here. Statements about the printed diagrams below refer only to the preprint.

Acquired inputs. The accepted broad-primitive enumeration exhausts all 95 subgroup anchors in a fixed permutation group G of order 20160, using the retained complete subgroup and maximal-inclusion inventories. Its full-automorphism fusion has 29 representatives of ranks 2–5. The separately accepted palette calculation, together with its exact serialization repair, provides complete right-coset partitions, incidence rows and actual G-actions for these representatives. The full-automorphism and GRAPE/nauty completeness premises remain explicit acquired inputs. They are not proved by checking positive permutation maps. The previously accepted projective construction component provides two rank-five models and eight literal truncation routes.

Theorem 1. *The retained literal calculations establish the following statements.*

1. *The numerical node data, all rank-two residues, local action kernels and local* 2*-transitivity flags are determined for each of the 29 acquired representatives. Within this complete catalogue, every preprint diagram has one representative matching its recorded invariants, subject to the two discrepancies in Lemma 1.*

2. *All 52 printed circular, Petersen and affine-plane bond labels have explicit canonical incidence maps, including their displayed polarity.*

3. *The accepted constructions labelled 5.1 and 5.2 are explicitly isomorphic to catalogue representatives 26 and 28, respectively. The eight acquired truncation routes also have complete object and incidence maps.*

This is not an assertion that diagrams determine geometries in general. Abstract names of Borel groups, parabolics, kernels, automorphism groups and correlation groups require their own structural identifications.

Proof. Choose the identity-coset chamber. For a set J of remaining types, intersect the neighbour sets of all complementary chamber objects. These are exactly the objects of the base residue of type J. Acquired flag transitivity transports this calculation to every flag of the same type. A panel order is its cardinality minus one. In its actual stabilizer, the stabilizer of one panel object is tested for transitivity on the other objects. No local 2-transitivity assumption is imposed.

For every pair of types, all incidence and nonincidence rows of the resulting bipartite graph are retained. Breadth-first distances give eccentricities, compared at both ordered endpoints of each printed numerical $a/g/b$ label, and the minimum cycle length; half that length is the gonality. A residue is a digon precisely when it is complete bipartite. At each base object, every element of its parabolic is tested on every object of its residue. The elements fixing all these objects are exactly the local action kernel. Only its order, not its abstract structure, is used in the numerical join.

For a circular bond, the degree-two objects map bijectively to all unordered pairs of the opposite object set. For a Petersen bond, an explicit map sends the ten points to the two-subsets of a five-set and the fifteen lines to disjoint pairs of such subsets. Every incidence row is checked. For an affine bond, sixteen points and twenty four-point lines have a unique line through each point pair and five parallel classes. Adding one point for each parallel class and a line at infinity gives a projective completion. An explicit 21-point bijection takes all its lines to the coordinate plane over $\mathbb{F}_4$. The recorded point/line maps also verify the source polarity.

Finally, the transcription records actual node positions, not presumed type numbers. All permutations compatible with the printed type sizes are checked against the complete residue data. The resulting table below records the unique catalogue index in each case; symmetric node permutations may remain. These

tests compare the complete finite catalogue with the source, rather than using source counts as acceptance criteria. □

Source rank 2	2.1	2.2	2.3	2.4	2.5	2.6	2.7				
Catalogue	1	4	2	5	3	6	7				
Source rank 3	3.1	3.2	3.3	3.4	3.5	3.6	3.7				
Catalogue	9	13	10	11	14	12	8				
Source rank 4	4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	4.10	4.11
Catalogue	15	24	19	22	23	18	17	16	21	20	25
Source rank 5	5.1	5.2	5.3	5.4							
Catalogue	26	28	29	27							

Construction maps and action conventions. The projective component and census use different generator lists. Their complete generated permutation groups are verified equal on the same 21 labels. The accepted coordinate-to-retained map is reused. If necessary, an explicit duality is applied: a normal vector labels its perpendicular projective line, and the induced action on these 21 lines is transported back to the fixed point labels. This is checked to permute all 20160 group elements bijectively and preserve every edge of the generator Cayley graph, so it is an automorphism α of the actual group.

For every proposed object x_i, the generators of the actual parabolic H_i fix x_i after applying α. Consequently

$$H_i g \longmapsto x_i^{\alpha(g)}$$

is well defined. The certificate additionally checks every coset member, the complete object bijections, all incidence and nonincidence rows, and the explicit type permutation. Separate checks of the saved maps verify 3,185 incidence rows and 6,370 actual generator-action rows.

Lemma 1. *In preprint diagrams 5.1 and 5.2, the printed order* 2 *at the unique 210-object node must be* 1 *for the stated constructions.*

Proof. The literal construction maps send the 210-object node to the unordered point-pair type. Its panel has two objects and therefore order 1, as already verified in the accepted construction. The order-2 line node has 21 objects and cannot be confused with this node.

□

Scope and reproducibility. The scripts retain all 29 literal residue tables, 52 canonical bond maps and ten construction/truncation maps. Bounded controls include 25 known graph distance/girth examples, six complete-graph incidence examples, damaged maps and missing endpoints. Two failed preliminary joins are preserved: one incorrectly required identical generator lists, and one omitted the needed duality. Neither negative attempt is a premise. No graph-isomorphism target or subgroup census is repeated by this component.

The original preprint has a single horizontal bond at the base of diagram 3.5, consistent with the literal six-cycle residue. The first author edition misread this bond and incorrectly proposed a source correction; that edition and its passing finite checks are preserved, and this false claim is withdrawn. The successor additionally compares both ordered endpoint eccentricities in every numerical $a/g/b$ bond label, not only its middle gonality. The source has no local-2-transitivity marker at 3.5 or 4.11, consistent with the actual panels. Earlier provisional transcription mistakes at those two rows are disclosed and corrected. The two discrepancies above do not establish an error in the uncollated journal edition. Remaining constructions, the general doubling hypotheses, abstract group-name claims and whole-source completion are outside this component. There is no new question, Lean, measurement or novelty claim.

A Baer-subplane construction of Geometry 3.3 and a correction to its printed incidence rule

L100 finite construction component

We use the fixed 22-page Halle preprint of Gottschalk and Leemans [1], physical page 19. Its later journal edition has not been collated. Put $P = \mathrm{PG}(2,4)$ and $G = \mathrm{PSL}_3(4)$. The coordinate plane, its exact identification with the retained 21-point action, and the complete coset representative numbered 10 are acquired from the accepted projective-construction, palette and source-join components. The latter representative is Geometry 3.3.

Theorem 1. *For each of the three G-orbits of Baer subplanes, the three incidence rules printed for Geometry 3.3 admit a two-object flag which does not extend to a chamber. Retain the first two rules and restrict the third to pairs having a common incident Baer subplane in the selected orbit. Each of the three resulting incidence systems is isomorphic, with its specified G-action transported by an explicit automorphism of G, to the acquired Geometry 3.3. The printed and corrected line-pair to point-pair degrees are respectively* 18 *and* 6.

This corrects an ancillary construction recipe. It does not refute the paper's principal classification, which already contains the corrected geometry. Other constructions and the complete source disposition are outside this component.

1 Complete subplane input

Lemma 1. *A seven-point subset B of P is an embedded plane of order two if and only if every ambient line meets it in one or three points. There are exactly* 360 *such subsets.*

Proof. Under the intersection condition every pair of points lies on a unique three-point intersection. These give $\binom{7}{2}/\binom{3}{2} = 7$ lines, three through each point. Each such line meets the six others, twice at each of its three points, since two lines cannot share two points. Thus the induced incidence system is an embedded Fano plane. Conversely, a Fano subplane has seven ambient lines meeting it in three points. Any other ambient line meets it in at most one point. It cannot avoid the subplane: its intersections with the seven subplane lines would be distinct, although it has only five points.

An ordered projective frame is a sequence of four points with no three collinear. There are $21 \cdot 20 \cdot 16 \cdot 9 = 60480$ ambient frames. Every frame lies in a unique Fano subplane. Indeed a projectivity sends it to $[1,0,0], [0,1,0], [0,0,1], [1,1,1]$. The three remaining Fano points are forced by intersections of the joins of these four points; they are $[1,1,0], [1,0,1], [0,1,1]$. These seven points do form a subplane, so existence and uniqueness both hold. Each Fano plane has $7 \cdot 6 \cdot 4 \cdot 1 = 168$ ordered frames. Counting frames with their unique subplane gives $60480/168 = 360$. □

The preserved first finite pass examined all $\binom{21}{7} = 116280$ subsets and saved the 360 qualifying sets before failing its later, uncorrected incidence join. The current checker uses these saved sets: it verifies distinctness and the lemma's criterion, then closes each listed orbit under the actual acquired generators. The result is three orbits of size 120. Completeness follows from the lemma; it is not inferred from a printed orbit count. No subset search is repeated.

2 The printed rule and its repair

For one orbit $\mathcal{B}$, the types are $\mathcal{B}$, unordered pairs of distinct lines, and unordered pairs of distinct points. Write pq for the line joining p and q. The printed rules are

$$B * \{\ell, m\} \iff B \cap (\ell \cup m) = \ell \cap m, \tag{A}$$

$$B * \{p, q\} \iff B \cap \{p, q\} = \varnothing \text{ and } |B \cap pq| = 3, \tag{B}$$

$$\{\ell, m\} * \{p, q\} \iff (\ell \cup m) \cap \{p, q\} = \varnothing \text{ and } \ell \cap m \cap pq \neq \varnothing. \tag{C}$$

These are literal transcriptions of the native source, not of its OCR. Let $o = \ell \cap m$. In (C), pq must be one of the three other lines through o. On each there are four points outside $\ell \cup m$. Consequently (C) has degree $3\binom{4}{2} = 18$ at every line pair.

Here is an explicit finite obstruction to flag extension. Label the retained points $0, \ldots, 20$ as in the acquired coordinate-to-permutation map, and encode a point set X by $\sum_{x \in X} 2^x$. Fix $\ell = \{0, 1, 2, 3, 4\}$ and $m = \{0, 5, 6, 7, 8\}$. In the three orbits ordered by their least encoded subplane, the following table gives a printed incident point pair and the complete two sets of Baer neighbours.

Orbit	Point pair	Neighbours of $\{\ell, m\}$	Neighbours of the point pair
0	$\{9, 11\}$	710657, 730625, 1361409, 1390593	46093, 87059, 398529, 1578273
1	$\{9, 10\}$	816129, 887297, 1206785, 1283073	47297, 88353, 399373, 1579027
2	$\{9, 10\}$	438785, 478209, 1616897, 1659393	30741, 104459, 792737, 1186113

The joining line in each case is $\{0, 9, 10, 11, 12\}$, so (C) holds. The two neighbour sets in each row are disjoint. The checker obtains each list by applying (A) and (B) to every member of that orbit, using an independent set formulation. Thus each displayed pair is a flag with empty remaining-type residue. This proves the stated defect of the printed recipe.

The repaired relation is

$$\{\ell, m\} *' \{p, q\} \iff \text{(C) holds and there exists } B \in \mathcal{B} \text{ satisfying (A) and (B).}$$

Rules (A) and (B) are unchanged. The selected orbit is essential. For additional exact context, the 3780 printed pairs split into three sets of 1260: in each pair precisely one orbit supplies common Baer neighbours, and it supplies two. The checker verifies this finite partition directly. In each repaired system the degree is six.

3 Literal positive identifications

The acquired catalogue representative has maximal parabolics H_i and objects the right cosets $H_i g$. For each of the three repaired systems the certificate supplies a type permutation, base objects x_i and a point permutation h. It checks that h preserves the actual lines and conjugates the full, explicit 20160-element group onto itself. With $\alpha(g) = h^{-1} g h$, the object map is

$$H_i g \longmapsto x_i^{\alpha(g)}.$$

The literal parabolic generators fix x_i. Moreover the verifier checks all 60480 coset-member assignments in each map, so it verifies well-definedness independently of the choice of coset representative. It then checks each typewise bijection, all 189000 directed cross-type object pairs, and all 1080 generator-object action pairs. Line pairs are pairs of five-point sets acted on by actual point permutations; no point-label/line-label identification is assumed.

All three maps pass. Therefore incidence, nonincidence and the specified action agree exactly. The already proved geometry, connectedness, residual properties, flag transitivity and RWPRI properties transport through these maps. In particular the three acquired local two-transitivity flags are all true. Equivalently, composing one map with the inverse of another gives a full incidence equivalence between any two orbit choices; this is not inferred from their equal sizes.

The certificate's finite verification, its exact acquired input interfaces and the short lemmas above constitute the proof. The acquired group order and full coset catalogue retain their previously stated computational and classification premises. The positive maps do not require a new group, subgroup, automorphism or graph-isomorphism search. The first failed join, the prelaunch ordering correction and the changed successful join are preserved separately. This component makes no novelty, measurement, open-question or whole-source claim.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL$(3, 4)$, retained Halle preprint 99–30 (1999), physical page 19. The later journal edition is not collated here.

Doubling, truncation and fusion joins for the $\mathrm{PSL}_3(4)$ catalogue

L100 finite source-join component

We use the fixed Halle preprint of Gottschalk–Leemans [1]; the later journal edition remains uncollated. Its labels $2.1, \ldots, 5.4$ are joined to the accepted 29 literal coset representatives by the previous source-join component. The complete subgroup inputs, broad primitive enumeration, full-automorphism fusion and palette calculations retain their exact previously stated premises. In particular local two-transitivity was tested, not imposed as an enumeration filter.

Theorem 1. *The four printed doubling links are valid:*

$$2.7 \longrightarrow 2.6, \quad 4.1 \longrightarrow 4.8, \quad 5.2 \longrightarrow 5.3, \quad 5.1 \longrightarrow 5.4.$$

Of the 24 printed truncation links, 23 have complete literal incidence maps. The remaining assertion, on physical page 9, that 3.6 *is a truncation of* 5.3 *is false by type cardinalities; replacing* 5.3 *by* 5.2 *gives a verified truncation. All 29 printed parenthetical fusion multiplicities agree with the complete acquired 145-family partition.*

The truncation error concerns an ancillary link. It does not refute the principal classification. This component does not identify the abstract names of parabolics, kernels, automorphism or correlation groups.

1 A concrete doubling argument

Lemma 1. *Let Γ be a flag-transitive geometry with point type X and another type E. Suppose the X-shadows of the objects of E are distinct two-element sets. Assume that, for every object z of any other type,*

$$e * z \quad \Longleftrightarrow \quad \sigma_X(e) \subseteq \sigma_X(z). \tag{1}$$

Replace X, E by copies X^+, X^- of X. Declare $x^+ * y^-$ when $\{x, y\} = \sigma_X(e)$ for some $e \in E$; declare either copy of x incident with z exactly when $x * z$ in Γ; leave all other incidences unchanged. The resulting incidence system is a geometry, and the induced specified group action is flag-transitive.

Proof. A flag using both copies gives distinct points x, y and a unique object e with shadow $\{x, y\}$. By (1), replacing y^- by e gives an old flag through x, e and the unchanged objects. Extend it to an old chamber. Its associated new chamber extends the original flag. A flag using only X^+ extends from the corresponding old flag through x; the other point of its chamber's E-object supplies the minus copy. With only X^- interchange these roles. With neither, extend the unchanged old flag first. Thus every new flag extends.

The chamber bijection sends (x, e, Z) to (x^+, y^-, Z). Here y is the other point of e, and the bijection is equivariant. Consequently old chamber transitivity implies new chamber transitivity; in a geometry this implies flag transitivity by extending flags to chambers. The proof works in rank two as well. □

Condition (1) also implies the particular (I_{12}) condition printed on page 6. For an object e and an object z of another type, their point-shadow intersection is empty, a singleton $\{x\}$, or the full shadow of e. In the singleton case the flag $\{x\}$ is incident with both and has the required point shadow. In the full case (1) gives $e * z$, so the flag $\{e\}$ works. Incidence here is reflexive, as in the source's coset definition.

The source's quoted Corollary 2.1 is explicitly a rank-at-least-three statement attributed to its reference [40]. Its use at 2.7 $\to$ 2.6 therefore requires a qualification. The direct rank-two argument above supplies that case: 2.7 is the edge-incidence system of a graph, and the construction is its bipartite double. We do not extend the scope of the quoted corollary by assertion.

For $4.1, 5.1, 5.2$, the accepted projective model supplies ordinary points and unordered point pairs. Incidence with a hyperoval means containment, and incidence with the relevant line means disjointness, for both the point and pair types. Thus (1) also has this immediate projective explanation. Independently, the finite verifier checks every pair shadow, its uniqueness, and every instance of (1) in the actual coset models. No source point or line label is assumed to equal a coset label.

2 Exact identifications and all truncations

All new maps use the same positive certificate interface. If H_i is an actual target parabolic, the certificate gives a derived source object x_i fixed by its images under an explicitly checked group automorphism α. On target right cosets the map is

$$H_i g \longmapsto x_i^{\alpha(g)}.$$

The derived geometry is either a doubled model from the lemma or an induced system on a specified subset of source types. The original coset actions are checked against the saved generator permutations; right multiplication, not a guessed handedness, defines all object images. Allowed automorphisms are literal conjugations by acquired point normalizers and the acquired coordinate duality. Their exact generator images and full group bijections are retained.

For every map the verifier checks all coset members, all typewise bijections, every incidence and nonincidence row, and every generator action on every object. Thus the target's full geometry and residue properties transport, including its actual local two-transitivity flags. This argument does not infer a map from matching group orders, diagrams or object counts. The four doubling maps require respectively $40320, 80640, 100800, 100800$ coset-member checks.

The complete positive truncation ledger is as follows. Eight links were already accepted in the projective-construction component and are acquired unchanged; sixteen additional maps are checked here.

Target	Verified parent geometries
2.2	5.1, 5.2, 4.10, 3.1, 3.4, 3.6
2.4	4.11, 3.5
2.5	3.3
3.1	5.1, 5.2, 4.1
3.2	4.3, 4.4
3.6	5.1, **5.2**, 4.1, 4.10
4.1	5.1, 5.2
4.8	5.3, 5.4
4.9	5.2
4.10	5.1

The bold entry repairs the printed 5.3. Indeed, a truncation retains the complete object sets of selected types. The type sizes of 3.6 are $56, 210, 56$, whereas those of 5.3 are $21, 21, 21, 56, 56$; these are also visible directly in the native diagrams on pages 9 and 13. No truncation of 5.3

can introduce a 210-object type. The replacement 5.2 is established positively by its saved map. It is also consistent with the projective constructions: 5.1 and 5.2 differ only in a line–hyperoval relation, whereas this truncation keeps the two hyperoval types and the point-pair type.

3 Fusion multiplicities and evidence boundary

The acquired complete fusion records contain 145 distinct inner-conjugacy families with type permutations allowed, partitioned by the actual full $\mathrm{Aut}(G)$ action. The rank totals are $21, 25, 63, 36$ for ranks $2, 3, 4, 5$. Each retained family occurs once. The 29 representative identities agree exactly with the independently acquired bare catalogue. Grouping the saved membership records by these exact identities gives the following source-order multiplicities:

Rank	Multiplicities in increasing source-label order
2	1,6,3,3,6,1,1
3	6,3,3,3,3,6,1
4	6,3,3,6,6,6,3,6,6,12,6
5	12,6,6,12

These agree with all parenthetical numbers on pages 7–13. The families are counted modulo G-conjugacy and type permutations, then fused under $\mathrm{Aut}(G)$, not extra bare geometries or correlation automorphisms. The projection parser accepts only the complete restricted membership-record syntax; it checks all 145 distinct member identities and all 29 representative joins. The completeness and inequivalence of those records are acquired from the accepted enumeration and fusion proofs, not re-proved by counting their rows.

The first truncation pass saved ten complete positive maps before rejecting the impossible printed link. That failed terminal receipt is preserved. A changed pass handled only the six outstanding positive maps, including the explicit replacement. Default replay verifies all twenty new maps directly and does not repeat candidate-map discovery, an ambient census or a graph-isomorphism search. Small-model controls check the doubling construction and chamber correspondence; separate damaged maps and record controls enforce the exact interfaces. Abstract structural names, ordered-butterfly and oval constructions, and the final whole-source synthesis remain separate. There is no new measurement, Lean, question or priority claim.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* $\mathrm{PSL}(3, 4)$, Halle Reports on Algebra and Geometry 99–30 (1999), fixed 22-page preprint.

Ordered butterflies and oval constructions for $\mathrm{PSL}_3(4)$

L 100 finite construction certificate

We certify seven concrete constructions in the fixed 22-page Halle preprint of Gottschalk and Leemans [1], using the already certified point action of $G = \mathrm{PSL}_3(4)$ and its coset geometries. The reference is the preprint, not an independently collated journal edition. We make the butterfly convention and the otherwise unspecified oval classes explicit. This is a construction component, not a new classification or an assertion that all remaining structural names in the source have been proved.

Acquired finite premises. The coordinate model is the plane over $\mathbb{F}_2[t]/(t^2 + t + 1)$, with an explicit bijection from its normalized coordinate vectors to the retained 21-point permutation model. Its saved 20160 permutations and two generators are the same literal group as the coset catalogue. The acquired point/line/hyperoval certificate gives the 21 lines and three 56-element hyperoval orbits. The acquired Baer certificate gives the three 120-element orbits $\mathcal{B}_0, \mathcal{B}_1, \mathcal{B}_2$ of Fano subplanes. Completeness of these 360 subplanes follows from the frame count $21 \cdot 20 \cdot 16 \cdot 9/(7 \cdot 6 \cdot 4) = 360$, with a unique embedded Fano plane through a frame. These literal inputs, their proofs and their distinct reviews are acquired without repeating a group or subgroup census.

An explicit convention. An ordered butterfly in this note is an ordered pair of unordered poles $b = (\{B_1, B_2\}, \{C_1, C_2\})$, consisting of four distinct subplanes, such that, for a point p and a line z,

$$\begin{aligned} B_i \cap C_j = \{p\} \quad (1 \le i, j \le 2), \qquad T_+ = B_1 \cap B_2, \quad T_- = C_1 \cap C_2, \\ |T_+| = |T_-| = 3, \qquad T_+ \cup T_- = z. \end{aligned} \tag{1}$$

This all-cross-intersections convention is stronger than the two displayed cross-intersection conditions on source page 4; the distinction is addressed on the last page. Write $U_+ = B_1 \cup B_2$, $U_- = C_1 \cup C_2$ and

$$Q_+ = T_+ \setminus \{p\}, \qquad L_\pm = \{l : p \in l \subseteq U_\pm\}, \qquad H_+ = \{(B_i \setminus z) \cup (T_- \setminus \{p\}) : i = 1, 2\},$$

with H_- defined by interchanging the poles. Thus a line "contained in a pole" means containment in the union of its two point sets. The central line is not such a line: its intersection with that union has only three points.

Lemma 1. *For each $\mathcal{B}_i$, (1) gives exactly 1260 ordered butterflies, one orbit of G, with stabilizer $D_8 \times C_2$. Both L_+ and L_- have two lines. The four members of $H_+ \cup H_-$ are distinct hyperovals in one orbit $\mathcal{H}_{e(i)}$, and $\mathcal{B}_i$ meets every hyperoval in this orbit evenly. The positive pole determines the ordered butterfly uniquely. Its unordered four-subplane set is also determined by an opposite pair of subplanes or by one of its subplanes together with (p, z).*

Proof. The finite certificate has a complete, small domain. Test all $\binom{360}{2} = 64620$ subplane pairs. Retain a pair when its intersection is a collinear triple T. There are 3780 such pairs. For each, its line z is unique. For each of the three choices $p \in T$, test all retained pairs whose intersection is $(z \setminus T) \cup \{p\}$, and impose all four equalities in (1). This exhausts every possible opposite pole. The combined search over these three choices gives exactly one opposite pole for each retained pair. The resulting objects have 1260 members in each Baer orbit. Closure and transitivity are

checked using the two actual group generators. The alternative uniqueness assertions are checked by grouping the full list by the indicated exact subplane/flag keys, never by numerical invariants.

For a representative in each orbit, test all saved group permutations for stabilization of the ordered poles. The 16 retained permutations are verified closed. The certificate exhibits a, b, z with $|a| = 4$, $|b| = |z| = 2$, $bab = a^{-1}$ and $[a, z] = [b, z] = 1$; its sixteen words $a^i b^j z^k$, $0 \le i < 4$, $0 \le j, k < 2$, are distinct and exhaust the stabilizer. This proves its asserted structure, not merely its order. The source's auxiliary subplane-stabilizer transitivity is checked on all 21 partners of one subplane in each orbit.

Each asserted hyperoval is checked against the complete acquired list. All $168\binom{6}{4} = 2520$ four-subsets of these hyperovals are distinct; therefore the associated quadrangle has a unique such completion. Containment tests give the two pole-line sets, the even-intersection assertion and all stated uniqueness keys. These are finite set tests on the complete domains just described; the replay recomputes them. □

The six butterfly geometries. For $B \in \mathcal{B}_i$, set $R(B, b) \iff T_+ \subseteq B$ and $B \notin \{B_1, B_2\}$. The following table states every incidence relation between distinct types. Here h, h' range over two tagged copies of the same class $\mathcal{H}_{e(i)}$, x is a point, l a line, and b an ordered butterfly from $\mathcal{B}_i$. The ordered types and the six entries are respectively the relations 12,13,14,23,24,34. Equal-type incidence is equality.

Source	Types	Relations in the stated order
4.2	l, x, h, b	$x \notin l;\ l \cap h = \varnothing;\ l \in L_+;\ x \in h;\ x \in Q_+;\ h \in H_-$
4.3	h, h', B, b	$h \cap h' = \varnothing;\ \lvert h \cap B\rvert = 4;\ h \in H_-;$
		$h' \cap B = \varnothing;\ h' \in H_+;\ R(B, b)$
4.4	h, h', x, b	$h \cap h' = \varnothing;\ x \in h;\ h \in H_-;\ x \notin h';\ h' \in H_+;\ x \in Q_+$
4.5	l, x, h, b	$x \notin l;\ l \cap h \neq \varnothing;\ l \in L_-;\ x \in h;\ x \in Q_+;\ h \in H_-$
4.6	x, h, B, b	$x \in h;\ x \in B;\ x \in Q_+;\ \lvert h \cap B\rvert = 4;\ h \in H_-;\ R(B, b)$
4.7	l, x, B, b	$x \notin l;\ \lvert l \cap B\rvert = 3;\ l \in L_+;\ x \in B;\ x \in Q_+;\ B \in \{B_1, B_2\}$

These are the native signs on pages 16–18. In particular 4.5 uses type-one hyperovals from 4.4 and the negative pole for its line relation.

Oval and dual-oval classes. Use the coordinate polarity $[a] \mapsto \{[v] : a \cdot v = 0\}$ with its literal transport to the saved point and line labels. It induces a permutation d of the three Baer classes. For each hyperoval orbit, delete each of the six points from each hyperoval. This gives 336 distinct ovals in one orbit; the deleted point is its nucleus, since the five tangent lines are checked to pass through it. Polarity gives the corresponding dual ovals, with a nucleus line. For each ordered pair $i \neq j$, choose

$$B \in \mathcal{B}_i, \quad C \in \mathcal{B}_j, \quad O \in \mathrm{Ovals}(\mathcal{H}_{e(i)}), \quad D \in \mathrm{DualOvals}(\mathcal{H}_{e(d(j))}). \tag{2}$$

In the saved class ordering, $e = (0, 1, 2)$ and $d = (0, 2, 1)$. The indices are part of the explicit model, not intrinsic names. Writing $n(O)$ for a point and $n(D)$ for a line, the source 4.11 relations are

$$\begin{aligned}
B \sim C &\iff B \cap C = \varnothing,\\
B \sim O &\iff |B \cap O| = 4,\\
B \sim D &\iff |B \cap n(D)| = 3 \text{ and } |B \cap l| = 1 \ (l \in D),\\
C \sim O &\iff n(O) \in C \text{ and } C \cap O = \varnothing,\\
C \sim D &\iff \#\{l \in D : |C \cap l| = 3\} = 4,\\
O \sim D &\iff |O \cap n(D)| = 2 \text{ and } (O \setminus n(D)) \cap l = \varnothing \ (l \in D).
\end{aligned}$$

All six choices in (2) are certified. The source names an oval class and a dual-oval class without specifying their relative identification; (2) supplies a successful precise choice, not a claim about arbitrary choices.

Theorem 2. *For each of the three Baer classes, the six butterfly recipes above are equivariantly isomorphic, up to an explicitly given automorphism of G and type permutation, to source geometries 4.2–4.7. All six choices in (2) similarly give geometry 4.11. Consequently these constructions are firm, residually connected, satisfy $(IP)_2$, and have a flag-transitive RWPRI G-action, with the local action qualifications stated below.*

Proof. We use the accepted coset catalogue, not a new recognition by orders or diagrams. The source-to-catalogue indices are respectively $24, 19, 22, 23, 18, 17, 25$. For each of the 24 maps, the certificate records a type permutation, one base object fixed by each actual parabolic, an explicit group automorphism α, and every object image. The map on right cosets is $P_t g \mapsto X_t^{\alpha(g)}$. Every one of the 20160 group entries in each of the four coset partitions is tested: all members of a coset give the recorded object. The recorded maps are bijections on each type. For every cross-type pair, incidence and nonincidence agree exactly; the two generator actions agree on every object. Thus these are actual typed incidence and action isomorphisms.

The automorphisms used are explicit point normalizers and coordinate duality. Point normalizers are checked on the whole saved group; duality is checked on every generator edge of its Cayley graph. The actual coordinate-to-retained label bijection is always used. None of these maps is inferred from a name, group order, or an unlabelled graph comparison.

The cited geometric properties, including extension of every partial flag, are acquired for these exact coset geometries. Isomorphisms transport all flags, chambers, residues and stabilizer actions, proving the assertions for the concrete constructions. Local double transitivity is not used as a recognition filter: in the displayed source type order,4.2 has flags $(1, 1, 1, 0)$,4.11 has $(1, 1, 0, 0)$, and 4.3–4.7 have four true flags. These follow from the acquired literal panel actions and the recorded type maps. □

A narrow qualification to the printed butterfly description. On page 4 the source displays only $B_1 \cap B_1' = B_2 \cap B_2' = \{p\}$, together with the triple/line condition. Those two displayed requirements alone do not imply the next sentence's uniqueness from (B_1, B_2). Here is a direct witness in the retained point labels. Fix

$$B_1 = \{1, 2, 3, 6, 7, 10, 11\}, \qquad B_2 = \{1, 4, 5, 8, 9, 10, 11\}.$$

Two different completions, in the same Baer orbit as B_1, B_2, are

p	B_1'	B_2'
1	$\{1, 12, 13, 15, 16, 18, 20\}$	$\{1, 12, 13, 14, 17, 19, 21\}$
10	$\{4, 8, 10, 12, 13, 16, 19\}$	$\{3, 7, 10, 12, 13, 14, 18\}$.

Both have central line $\{1, 10, 11, 12, 13\}$ and satisfy exactly the two printed cross equalities. In the first, all four cross intersections have size 1; in the second their sizes, by rows, are $(1, 3)$ and $(3, 1)$. All six point sets are checked against the complete saved subplane list; actual permutations of G mapping B_1 to each one are supplied. Hence neither same-orbit restrictions nor forgetting the pole ordering remove the ambiguity. This establishes insufficiency of the displayed conditions, not a claim that the conventional butterflies in the source's cited literature have that broader meaning. We do not inspect or contradict that external convention here. Convention(1) makes the intended constructions and the stabilizer assertion completely explicit. No principal classification is refuted by this ancillary qualification.

Reproducibility and attribution. The packet retains the initial failed uniqueness assertion under the weaker displayed rule, its diagnosis, and the successful successor. Default replay checks finite object completeness and the 24 supplied maps; it does not search for maps, rerun a subgroup census or recompute graph automorphism groups. The names and recipes are due to Gottschalk–Leemans and their cited antecedents. This certificate gives literal construction joins and an explicit scope qualification, with no novelty, priority, measurement or whole-source claim.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL$(3, 4)$, Halle preprint 99–30, fixed 22-page source, pp. 4–5,16–19.

Literal automorphism and correlation groups for the $\mathrm{PSL}_3(4)$ geometries

L100 structural certificate

This note identifies the groups attached to the 29 geometries in the fixed 22-page Halle preprint of Gottschalk and Leemans [1]. It acquires the complete literal coset models, their proved source-label identifications, and their complete palette automorphism groups. It does not repeat the ambient subgroup or geometry census or the GRAPE/nauty computation. The completeness of the latter groups remains an explicit acquired computational premise. The journal edition has not been independently collated.

Notation and acquired actions. Put $G = \mathrm{PSL}_3(4)$, in the retained 21-point action of order 20160. Its two actual generators, all its elements, and their actions on every coset model are acquired literally. Each palette graph has the objects and one additional vertex per type; objects and palette vertices have separate colours. Its full colour-preserving automorphisms are precisely $\mathrm{Cor}(\Gamma)$; the subgroup fixing every palette vertex is $\mathrm{Aut}(\Gamma)$. These statements use the accepted palette lemma, rather than the uncoloured incidence graph. Both groups contain the faithful embedded G.

On coordinates over $\mathbb{F}_4$, let δ be conjugation by $\mathrm{diag}(\omega, 1, 1)$, σ the field map $a \mapsto a^2$, and τ the inverse-transpose map induced by ordinary projective polarity. The explicit coordinate-to-retained-point bijection is acquired. In particular τ uses the bilinear dot product, not a Hermitian polarity. The twelve maps $\delta^i\sigma^j\tau^k$ form an actual subgroup

$$O = \langle \delta, \sigma, \tau \rangle \cong S_3 \times C_2, \qquad 0 \le i < 3, \quad 0 \le j, k < 2.$$

Indeed, both involutions invert δ and commute with each other. The certificate checks the complete multiplication table on the two generators of G. It checks that all 241920 maps obtained by following one of these twelve maps by conjugation by an element of G have distinct generator-image pairs. Thus O intersects the inner group trivially. The separately acquired complete computation $|\mathrm{Aut}(G)| = 241920$ identifies this explicit subgroup with all of $\mathrm{Aut}(G)$; the order is not inferred from the geometry table.

Define the following precise names:

$$\begin{array}{lll} \Sigma = G \rtimes \langle \sigma \rangle, & T = G \rtimes \langle \tau \rangle, & U = G \rtimes \langle \sigma\tau \rangle, \\ \Gamma = G \rtimes \langle \delta, \sigma \rangle, & V = G \rtimes \langle \sigma, \tau \rangle, & A = \mathrm{Aut}(G). \end{array} \tag{1}$$

Here Σ is the field-extension meaning of $P\Sigma L_3(4)$, explicitly documented by the primary Magma handbook [2], $\Gamma = P\Gamma L_3(4)$, and V supplies the explicit meaning of the source's $P\Sigma L_3(4) : 2$ in this table. Outer subgroups conjugate by O give the same names, with an actual conjugating element recorded.

Lemma 1. *For every saved typed or correlation group H, the certificate determines its conjugation image in $\mathrm{Aut}(G)$, its kernel, and an explicit splitting. It proves $H \cong G \rtimes E$ or $C_2 \times (G \rtimes E)$, for the recorded subgroup $E \le O$.*

Proof. Choose the smallest object type. Traverse the already known generators simultaneously on G and on this type. The 20160 restricted permutations are distinct, so this action is faithful. For each actual palette generator q and each generator g of G, look up the restriction of $q^{-1}gq$.

Then reconstruct the resulting element's action using its saved generator word and check equality on *every* object and palette vertex. Consequently every q normalizes the embedded G; restricted equality alone is never used as the conclusion. Its two generator images have a unique decomposition into an inner map and one of the twelve coordinate maps. The small quotient multiplication table determines E exactly.

The conjugation image contains all inner maps and projects onto E, so it is the entire preimage $G \rtimes E$, of order $20160|E|$. The acquired complete order of H therefore gives its kernel order, which is one or two in these 58 instances. This use of order follows the explicit image identification; it does not identify a group by order.

Choose short words in the actual generators lifting every element of E. Correct each lift by its exact inner factor, giving q_e whose conjugation on G is the specified coordinate map e. All products $q_e q_f q_{ef}^{-1}$ are checked literally. Discrepancies between the original generators and these normalized lifts are also checked. In the kernel-two cases their nonidentity values exhibit a single permutation z. The certificate verifies $z^2 = 1$, $z \neq 1$, and commutation with every generator of H and G. These relations detect the whole kernel: otherwise the normalized lifts together with G would already express every original generator with trivial kernel, contradicting the acquired order.

Write the multiplication discrepancies as $z^{c(e,f)}$. Solve the finite binary system

$$b(e) + b(f) + b(ef) = c(e, f) \pmod 2.$$

The certificate then checks all products of the actual corrected permutations $s_e = q_e z^{b(e)}$, including $s_1 = 1$. Thus these permutations form an actual homomorphic section. The subgroup generated by G and the s_e has the faithful conjugation image $G \rtimes E$ and has trivial intersection with $\langle z \rangle$. Together they generate H, proving the direct product assertion. In particular a central kernel of size two is never assumed to split without this section certificate. □

Theorem 2. *With notation (1), the literal models have the following groups. All five displayed direct factors C_2 are certified by the preceding lemma.*

Source	Aut(Γ)	Cor(Γ)	Preprint page
2.1	Γ	A	7
2.2	Σ	Σ	7
2.3	V	V	7
2.4	U	V	7
2.5	Σ	Σ	8
2.6	A	$C_2 \times A$	8
2.7	A	A	8
3.1	Σ	Σ	8
3.2	V	$C_2 \times V$	8
3.3	Σ	V	8
3.4	Σ	V	9
3.5	U	V	9
3.6	G	Σ	9
3.7	Γ	A	9
4.1	G	Σ	9
4.2	Σ	V	10
4.3	Σ	V	10
4.4	Σ	Σ	10
4.5	Σ	Σ	10
4.6	Σ	Σ	11
4.7	Σ	V	11
4.8	G	$C_2 \times \Sigma$	11
4.9	G	Σ	11

4.10	G	G	12
4.11	G	T	12
5.1	G	G	12
5.2	G	Σ	13
5.3	G	$C_2 \times \Sigma$	13
5.4	G	$C_2 \times G$	13

Proof. Apply the preceding finite certificate to the exact 29 acquired model/group pairs, with source labels supplied by the separately accepted literal source joins. Every generator is checked, not a proposed generating subset of the complete palette group. The supplementary data retain all outer/inner image pairs, lift words, normalization elements, section permutations and central involutions. The five nontrivial kernels occur only in the correlation groups of 2.6, 3.2, 4.8, 5.3 and 5.4. No typed group has a nontrivial kernel. □

Two precise name corrections. The preprint prints $P\Sigma L_3(4)$ for both Aut(3.5) on page 9 and Cor(4.11) on page 12, as it also does for groups of field type elsewhere. With the explicit field convention in (1), the first two names require replacement by U and T, respectively. This is not merely a relabelling of abstractly isomorphic groups. The acquired 20160-element list contains 315 involutions of G. For an involutory coordinate automorphism a, the involutions in its outer coset correspond exactly to

$$\{h \in G : h\, a(h) = 1\}.$$

Literal tests over this complete finite set give 120, 280 and 336 for $a = \sigma, \sigma\tau, \tau$, respectively. Thus Σ, U, T have 435, 595 and 651 involutions and are pairwise nonisomorphic. Even if one leaves the source's ATLAS-notation convention uninterpreted, its identical printed name cannot denote all three groups consistently. The unqualified $L_3(4) : 2$ at 2.4 is instead refined to U, not declared erroneous. All group orders remain unchanged, and these auxiliary name corrections do not refute the principal geometry classification.

Scope and attribution. The computations and proof concern only the complete typed and correlation group names. Abstract Borel, maximal-parabolic and boxed-kernel names are separate obligations; so are the final retained-theorem and source-contract synthesis. Root suggested using explicit coordinate and type actions and keeping kernel/splitting distinct from order agreement; this implementation and finite argument are the present author's work. The earlier construction, source-join and palette proofs and their distinct reviews are acquired inputs. No novelty, new census, current-status, Lean, measurement or whole-source credit is asserted.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL(3, 4), Halle preprint 99–30, fixed 22-page edition, especially pp. 4 and 7–13.

[2] Magma handbook, *Permutation representations of linear groups*, ProjectiveSigmaLinearGroup/PSigmaL.
`https://magma.maths.usyd.edu.au/magma/handbook/text/713`.

Parabolics, Borels and residue kernels in the $\mathrm{PSL}_3(4)$ catalogue

L100 literal structural certificate

This note identifies the actual subgroups printed in the 29 diagrams of Gottschalk and Leemans' fixed Halle preprint [1], pp. 7–13. The acquired complete coset catalogue and its source-node bijections are inputs. We do not repeat the subgroup or geometry census. The retained 21-point realization of $G = \mathrm{PSL}_3(4)$, its order 20160, all its literal permutations, and its verified coordinate identification with $\mathrm{PG}(2, 4)$ are explicit premises. The journal edition is not collated here.

For each catalogue geometry, H_i is the saved stabilizer of its base object of type i, $B = \bigcap_i H_i$, and K_i is the already certified kernel of H_i on that object's entire residue. These are literal subsets of the same ordered list of elements of G. There are 99 parabolic positions, 99 kernel positions and 29 Borels. Identical subsets, rather than equal orders or isomorphic groups, reduce these 227 positions to 77 atoms. Every source position retains its exact atom identifier. The certificate first checks that a stored generating list generates exactly its asserted subset.

Lemma 1. *The actual point or line stabilizers have structure $2^4 : A_5$. The actual unordered point-pair or line-pair stabilizers used here have structure $2^4 : S_3$, and an incident point-line stabilizer has structure $2^4 : A_4$. Each colon is paid by an actual complement.*

Proof. Choose coordinates with the fixed point represented by e_1. An element of its stabilizer has an $\mathrm{SL}_3(4)$ representative $\left(\begin{smallmatrix} a & u \\ 0 & C \end{smallmatrix}\right)$ with $a \det C = 1$. Multiplication by the scalar a^{-1} is permitted because $a^3 = 1$. It gives a unique representative with upper-left entry 1 and lower block in $\mathrm{SL}_2(4)$. Consequently this group is the split extension of the additive row space $\mathbb{F}_4^2$ by $\mathrm{SL}_2(4)$. The latter acts faithfully on its five projective points and is A_5. The dual calculation gives the line stabilizer. A flag imposes one fixed point in this five-object action, yielding A_4. A pair imposes an unordered two-set, whose stabilizer in A_5 is S_3.

These descriptions are joined to each actual H_i without assuming that coordinate and permutation labels agree. The check finds its invariant point, line, flag, or unordered pair in the retained projective model. For a fixed line ℓ, it defines N as the elements fixing all five points of ℓ, and L as the stabilizer within H_i of a specified point outside ℓ. Dually it uses a pencil and a line not through its centre. It verifies $|N| = 16$, a commuting involution basis of length four, $N \lhd H_i$, $N \cap L = 1$, and every element has a unique expression nl. The faithful five-object action of L is checked on every element; it is the entire A_5, its specified point stabilizer, or its specified two-set stabilizer. This proves the claimed splitting and the actual action, in addition to the abstract group name. □

Lemma 2. *The other large atoms are A_6, A_5 and $L_3(2)$, with the indicated faithful geometric actions.*

Proof. For each order-360 atom the checker finds a retained invariant hyperoval and checks all 360 induced permutations of its six points. They are distinct and even, hence are precisely A_6. For each order-60 atom it checks either an invariant oval or an actual five-line orbit. The 60 induced permutations are distinct and even, proving A_5 also for the dual-oval stabilizers. A point action is not silently substituted for a line action.

For each order-168 atom it finds a retained Baer subplane, checks the seven points and its seven triple lines, and verifies the unique-line axiom on every point pair. Every recorded action preserves all seven lines and is faithful. An automorphism of the Fano plane is determined by

an ordered noncollinear triple: the other four points are recovered by line intersections. Thus its order is at most $7 \cdot 6 \cdot 4 = 168$. The exhibited 168 permutations are its full automorphism group $\mathrm{GL}_3(2) = \mathrm{PSL}_3(2)$, as asserted. □

The small structures. All remaining atoms have order at most 72. Their certificates use the following complete normal forms; no small-group identifier or order-only recognition is used. Here D_8 and Q_8 both have order eight. An elementary abelian atom has a commuting prime-order basis and every coordinate word exactly once; a cyclic atom has an element of full order. For A_4 and S_4, conjugation on the four actual cyclic subgroups of order three is faithful and gives respectively all 12 even or all 24 permutations. The groups S_3 and $7:3$ have actual generators a, b, orders $(3,2)$ or $(7,3)$, and $a^b = a^2$ or a^4 as appropriate; all words $a^i b^j$ occur exactly once.

For D_8 use $|a| = 4$, $|b| = 2$, $a^b = a^{-1}$. For Q_8 use $|a| = |b| = 4$, $a^2 = b^2$ and $a^b = a^{-1}$. In both cases the eight words $a^i b^j$ are checked. The atoms $2 \times D_8$ and $2 \times Q_8$ additionally exhibit an involution z outside this eight-element subgroup which commutes with both generators; all sixteen words are distinct and exhaust the atom.

The three groups $3 : S_3$, $3^2 : 4$ and $3^2 : Q_8$ have an explicitly normal translation group $N = \langle a, b \rangle \cong \mathbb{F}_3^2$. The saved complement has order 2, 4 or 8, intersects N trivially, and its products with the nine coordinate words exhaust the group. Its conjugation matrices on (a, b) are checked literally. For order 2 the action is $-I$. For order 4 its generator squares to $-I$; its minimal polynomial $X^2 + 1$ is irreducible over $\mathbb{F}_3$, giving the specified cyclic-four action. For order 8 the matrices are exactly

$$\{I, -I\} \ \cup \ \{M \in \mathrm{SL}_2(3) : \mathrm{tr}(M) = 0\}, \tag{1}$$

the standard quaternion subgroup. This identifies the semidirect product action as well as its factors. The inversion extension of 3^2 can also be written $3 : S_3$: choose one cyclic translation factor as the normal subgroup and the other together with the inverter as the S_3 complement.

Theorem 3. *All 29 Borel names and all 99 parabolic names in the fixed preprint agree with the actual catalogue groups. All boxed kernel structures agree as well. The outer colon in diagram 2.3 requires the nonsplit qualification in Lemma 4; no group or geometry count changes.*

Proof. The preceding witnesses certify all 77 literal atoms. Independently transcribed names at each printed node are compared through the accepted source-node bijection, including nodes of equal size. The Borel names are listed below; the full 99-node table, with each actual parabolic, kernel, atom and extension certificate, is exported as `source_names.json`. It is a literal finite join, not a new classification by subgroup orders.

Diagram	B	Diagram	B	Diagram	B
2.1	$2^4 : A_4$	3.1	A_4	4.1,4.8	3
2.2,2.5	S_4	3.2	4	4.2–4.7	2
2.3	$3^2 : 4$	3.3,3.4	D_8	4.9,4.10	2
2.4	$7 : 3$	3.5	3	4.11	1
2.6,2.7	Q_8	3.6	S_3	5.1–5.4	1
		3.7	A_4		

□

Lemma 4. *Every recorded residue kernel is checked inside its actual parabolic. The split, direct and nonsplit extensions have the following explicit interpretations.*

Proof. The acquired kernel is rechecked to be contained and normal. For each nontrivial kernel the checker constructs its entire coset partition inside H_i and the literal quotient multiplication table. Identical pairs (H_i, K_i) share one certificate; 61 such pairs cover all 99 positions. A trivial kernel uses the identity map, without manufacturing a new quotient computation.

In diagram 2.1 the boxed 2^4 is exactly the translation radical of Lemma 1. In diagrams 2.2, 2.5, 3.1, 3.3, 3.4 and 3.7 the boxed 2^2 is a specific subspace of that radical. Its exact basis, a vector-space complement, and the actual Levi action are retained. Thus the unparenthesized printed expression is paid in the form $(2^2 \times 2^2) : L$, with the action on the *whole* radical. We do not assert a central direct factor of H_i or an invariant complement merely from this typography. The checker explicitly records whether its chosen vector complement is invariant.

In diagram 2.7 the actual Q_8 kernel has a commuting involution complement, proving the displayed direct product. In 3.2 the actual cyclic-four kernel has an elementary-four complement, proving $4 : 2^2$. In 3.5 the actual cyclic-three kernel has a disjoint S_3 complement, proving $3 : S_3$. These conclusions use the specified kernels, not just some isomorphic normal subgroups.

In the central nodes of 4.2–4.7 the boxed involution is central and every square belongs to it. The quotient is elementary abelian of order eight. The extension is nonsplit: a complement would commute with the central kernel and make the whole group elementary abelian, whereas its exhibited dihedral factor has elements of order four. This is consistent with the source's dotted 2.2^3 notation.

Finally diagram 2.3 has $H = 3^2 : Q_8$ and the actual kernel $K = 3^2 : C_4$ of index two. Every element outside K is checked to have order four; in particular none is an involution. Hence there is no complement of order two. This also follows symbolically by projecting to Q_8: an involution maps to 1 or its unique central involution, both in the cyclic-four image of K. The preprint's displayed outer colon in $(3^2 : 4) : 2$ must therefore be read with a nonsplit extension dot in place of that split colon. The corrected assertion is $(3^2 : 4).2 = 3^2 : Q_8$ with the specified kernel. It is an auxiliary structural-notation correction, not a refutation of the principal geometry classification. □

Verification and scope. All products above are products of the actual retained permutations. A separate witness validator checks the saved maps, normal forms, geometric factors, quotient tables and actions without searching for new witnesses. Damaged generators, action tables, kernels, complements and source associations are rejected. The development record retains the initial syntax failure and the unsuccessful point-only search for the dual-oval atom; the final edition handles the five-line action. The complete ambient list, the coset models, the kernel meaning and the source-node transport remain acquired inputs, with their earlier proofs and distinct reviews. This note supplies their missing internal structural identifications, not another ambient census or a new measurement. Root's earlier suggestions concerning geometric stabilizers were advisory; this component's literal code and proof are authored by the publication-recovery agent and await distinct review.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL$(3, 4)$, Halle preprint 99–30, fixed 22-page source, pp. 4, 7–13.

[2] J. H. Conway et al., *Atlas of Finite Groups*, Oxford University Press, 1985. Notation antecedent cited by [1]; no Atlas subgroup classification is used to identify these atoms.

Intermediate flag stabilizers in Geometry 5.1

L100 literal structural supplement

The construction and incidence properties of Geometry 5.1 in the fixed Halle preprint of Gottschalk–Leemans [1] have already been verified on its literal projective model. Its flag orders alone do not prove the intermediate structural names used in Lemma 4.3. This supplement supplies those names, using the acquired ambient permutation list and the acquired structural-certificate lemmas. No subgroup or geometry census is repeated.

Write the source types as $1 = \ell$ (line), $2 = p$ (point), $3 = h_3$, $4 = h_4$ (hyperovals in distinct specified orbits), and $5 = \{p, q\}$. Use precisely the saved construction chamber, whose five objects are the point masks

$$(31, 32, 27808, 2740, 160).$$

These masks refer to the accepted 21-point model, not to untransported coordinate labels or to the differently ordered coset catalogue. For every nonempty $I \subseteq \{1, \dots, 5\}$ define

$$H_I = \{g \in G : \ g \text{ fixes each of these objects of a type in } I\}.$$

The checker computes each set by testing all the acquired 20160 literal permutations, and joins its order to the accepted flag record. Since the acquired construction is flag-transitive, every other flag of the same type has a conjugate stabilizer. Thus a certificate at this chamber pays the structural assertion for every flag of that type.

Theorem 1. *All intermediate group names in Lemma 4.3 of the fixed preprint agree with the literal construction. The complete size-two and size-three flag tables are as follows; D_{10} has order ten.*

I	H_I	I	H_I	I	H_I	I	H_I
12	A_5	13	A_5	14	S_4	15	2^3
23	A_5	24	A_5	25	$2^4 : 3$	34	$3^2 : 2$
35	S_4	45	S_4				
123	D_{10}	124	S_3	125	2^2	134	S_3
135	2^2	145	2^2	234	S_3	235	A_4
245	A_4	345	S_3				

Proof. For each actual set the certificate first gives generators whose literal closure is exactly that set. The acquired parabolic-structure methods then supply faithful alternating actions for A_5; conjugation on four actual cyclic-three subgroups for A_4, S_4; commuting binary bases for the elementary abelian groups; and complete semidirect normal forms for S_3 and $3^2 : 2$. The latter has inversion action on its elementary nine-subgroup and is the same action denoted $3 : S_3$ in the acquired certificate. All witnesses are checked anew on these particular sets; an equal order or an abstract name in another catalogue position is never used as an identification. The two additional methods needed here are established below. □

Lemma 2. *The actual H_{123} is D_{10}. The actual H_{25} is a split $2^4 : 3$ with fixed-point-free complement action on the nonidentity elements of its normal elementary abelian subgroup.*

Proof. In H_{123} the saved elements a, b have orders five and two, respectively, satisfy $a^b = a^{-1}$, and the ten words $a^i b^j$ $(0 \le i < 5,\ 0 \le j < 2)$ are pairwise distinct and exhaust the actual set. These relations and normal forms give mutually inverse homomorphisms with the dihedral group of order ten.

In H_{25} the elements of order at most two form a 16-element subgroup N. The saved four generators are commuting involutions and their sixteen binary coordinate words are distinct. Normality is checked by conjugation with the saved generators of H_{25}. The saved element r has order three. Its conjugation action on the four basis elements is recorded as an actual matrix over $\mathbb{F}_2$, and it fixes no nonzero coordinate vector. All 48 words nr^j, $n \in N$, $0 \le j < 3$, are distinct and exhaust H_{25}. Thus $N \cap \langle r \rangle = 1$ and the extension splits with exactly the recorded action. To identify this action abstractly, its minimal polynomial divides $X^3 - 1 = (X-1)(X^2+X+1)$. Absence of fixed vectors eliminates the first factor, so N is the direct sum of two copies of the irreducible two-dimensional $\mathbb{F}_2 C_3$ module. This specifies the semidirect product beyond a mere order comparison. □

The same certificate covers the five singleton stabilizers $2^4 : A_5, 2^4 : A_5, A_6, A_6, 2^4 : S_3$, the five four-object stabilizers (order two for $1234, 1235, 1245, 1345$, and order three for 2345), and the trivial chamber stabilizer. There are therefore 31 exact nonempty flag types. The singleton structures reuse the acquired geometric stabilizer argument; all cyclic and elementary structures have literal normal forms.

Verification and source boundary. `certify.py` retains every element set and structural witness. Its saved-witness validation recomputes the actions and normal forms; damaged generators, action matrices, words, type assignments and source names are rejected. The native source page was read directly. The formula $G_{p,q}$ in its proof refers to the subgroup fixing both points, equivalently the stabilizer of the incident point and the unordered pair. The source's $3^2 : 2$ is normalized to the acquired name $3 : S_3$ only with its explicitly verified inversion action.

This supplement makes no new source correction. Residual connectivity, weak residual primitivity, complete flag extension, the ambient linear realization and flag transitivity are acquired with their prior proofs and distinct reviews. They are not inferred merely from this table of groups. The journal edition and unrelated cited classifications are not recertified here.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL(3, 4), Halle preprint 99–30, fixed 22-page edition, p. 15, Lemmas 4.2–4.3.